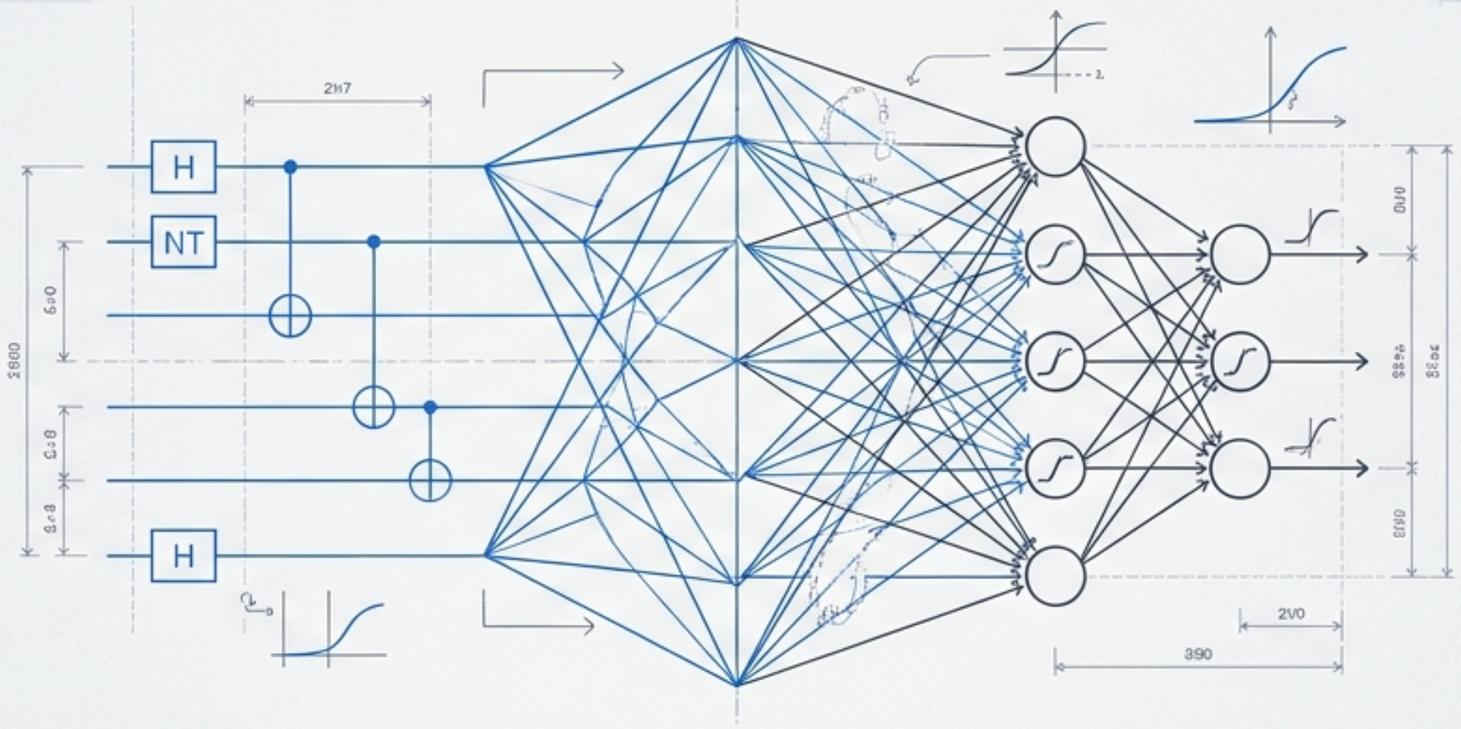
[Kubona: See, observe, discover — then know.]



An open-source platform for autonomous AI agents conducting quantum computing & verifiable scientific discovery.

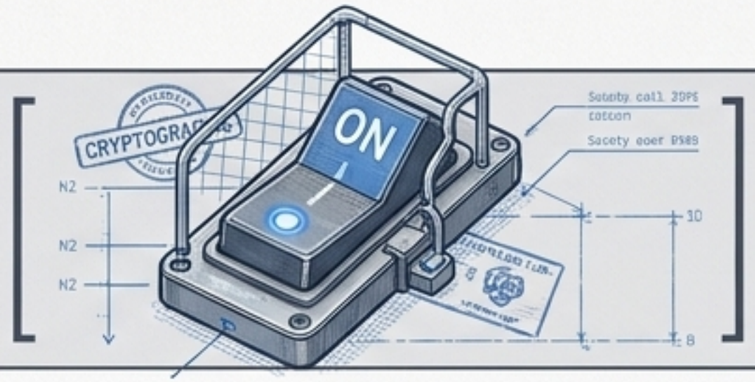
Origin: Chitonga (Tonga, Zambia) > *ku-bona* (to see, observe, perceive)

Gemini Notebook

[Dimension]	[Traditional Agentic AI]	[Kubona Epistemic AI]
Action Grounding	Relies on probabilistic guessing	37 deterministic tools via federated MCP stdio
Cost Constraints	Unbounded token consumption and silent API spend	Pre-flight budget checks & strict spend mandates
Failure Modes	Hallucinates plausible outputs and fails silently	Fails closed with plain-language, point-of-failure refusals
Result Verifiability	Unverified, ephemeral claims	Immutable SQLite WAL ledger and Merkle audit chains

Verified chain signature
Verified Chain Signature
Gemini Notebook

The 3 Immutable Design Principles



1. Receipts don't lie.

Every spend, evaluation, and agent action is written to a sealed ledger. Measurements are measurements. Guesses are labeled as guesses.



2. Refuse loudly.

No silent defaults. No invented numbers. The platform states its limitations at the exact point of failure.

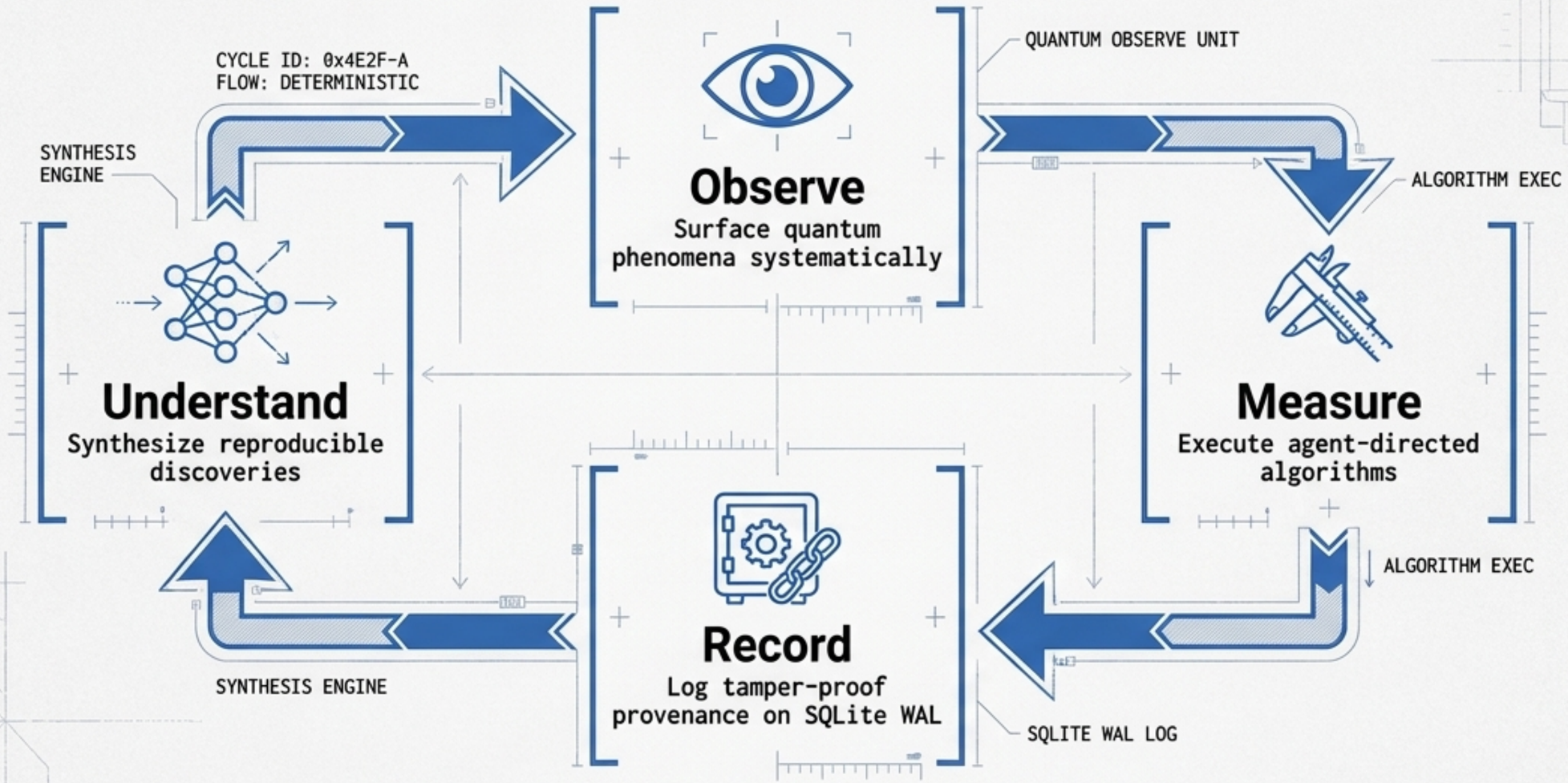


3. Count the tokens.

Every scaffold and agent loop must justify its token tax with a trust property. Underperforming scaffolding is removed.

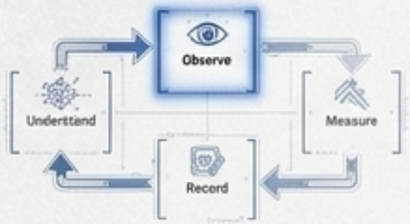
Verified Chain Signature
Gemini Notebook

The Epistemic Scientific Loop



A deterministic four-stage loop that forces AI agents to conduct science rather than generate approximations.

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Loop Stage 1: The Sovereign Observatory

[ATLAS__METERED]

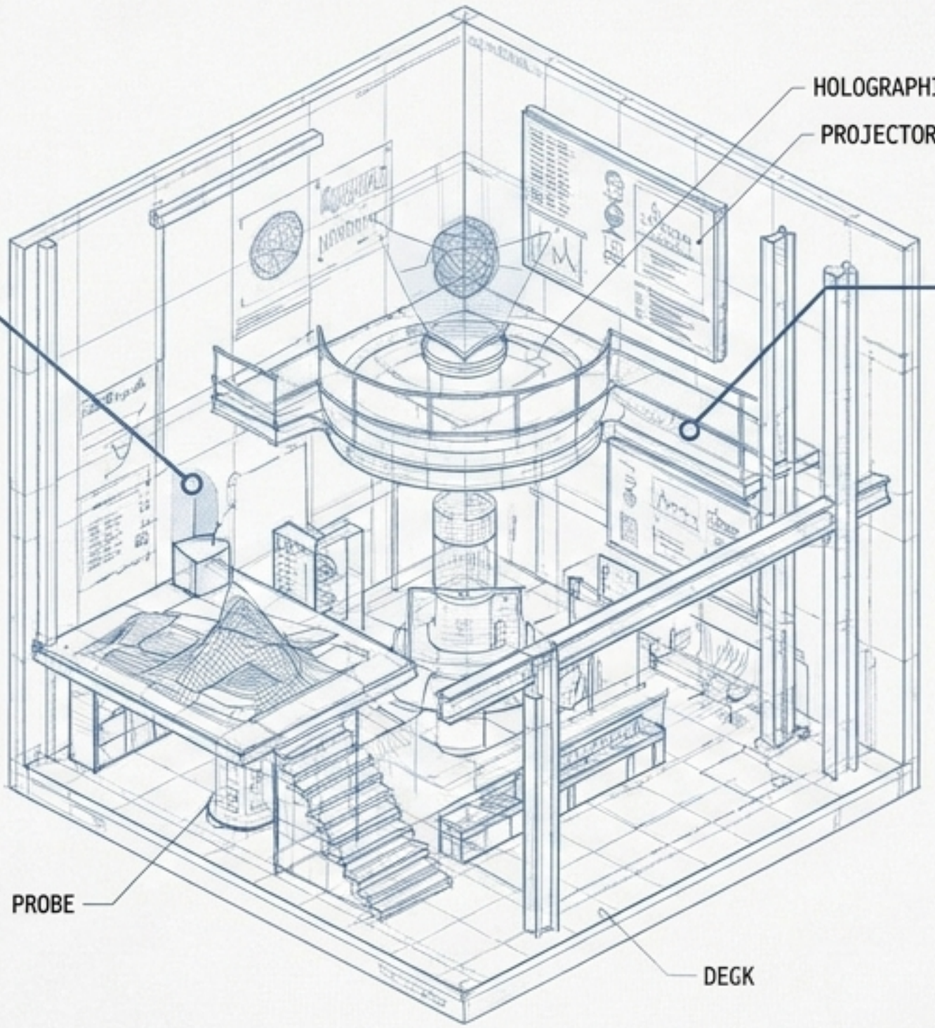
Tier 1: Walkable Lab

[$0.00 cost, air-gap safe.]

[LOCAL_EXEC]

- Local Three.js potential-energy terrain bound to the PySCF FCI anchor.
- Features a hover/click inspection probe and a calm-technology Attention Director.

[LOCAL_EXEC]
[PYSCF_FCI_ANCHOR]
[PROBE_V3.1]



[ATLAS_METERED]

Tier 2: Atlas Lens

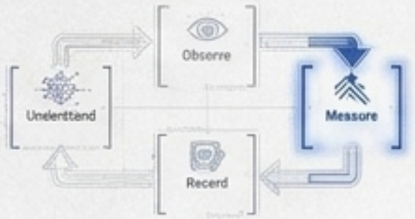
[Opt-in, metered ($0.50 reserve).]

[MARBLE_2_SKIN]

- World Labs Marble 2 architectural skin over local geometry.
- Snapshot posing, depth-conditioned chisel, and ledger-sealed promptUsed receipts.

[promptUsed]
[promptUsed_RECEIPT_ID]

Gemini Notebook



Loop Stage 2: The Quantum Bridge

[ATLAS__METERED]

Python Runtime

Agent execution generating VQE and QAOA algorithms.

[LOCAL_EXEC]
[PYSCF_FCI_ANCHOR]
[PROBE_V3.1]

Typed gRPC & Unix Domain Socket RPCs (unix://...)

[ATLAS_METERED]

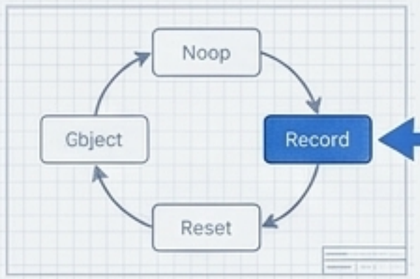
Quantum Core

Chemical anchors, circuit cutting, and hardware dispatch interfacing with PySCF, Qiskit Aer, and PennyLane.

[promptUsed]
[promptUsed_RECEIPT_ID]

Binding algorithms to physical and simulated QPU hardware across local and cloud environments.

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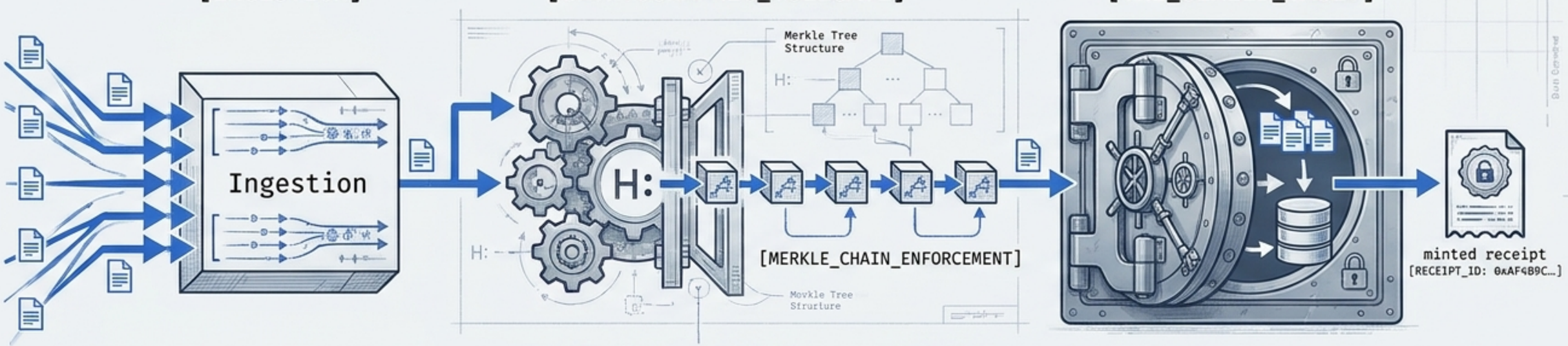


Loop Stage 3: Immutable Provenance

[INGESTION]

[CRYPTOGRAPHIC_TOLLGATE]

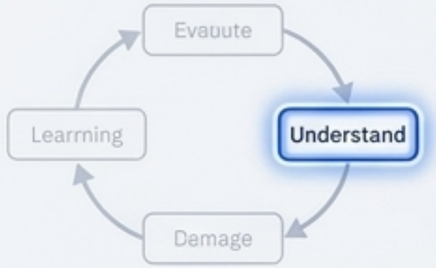
[THE_SEALED_VAULT]



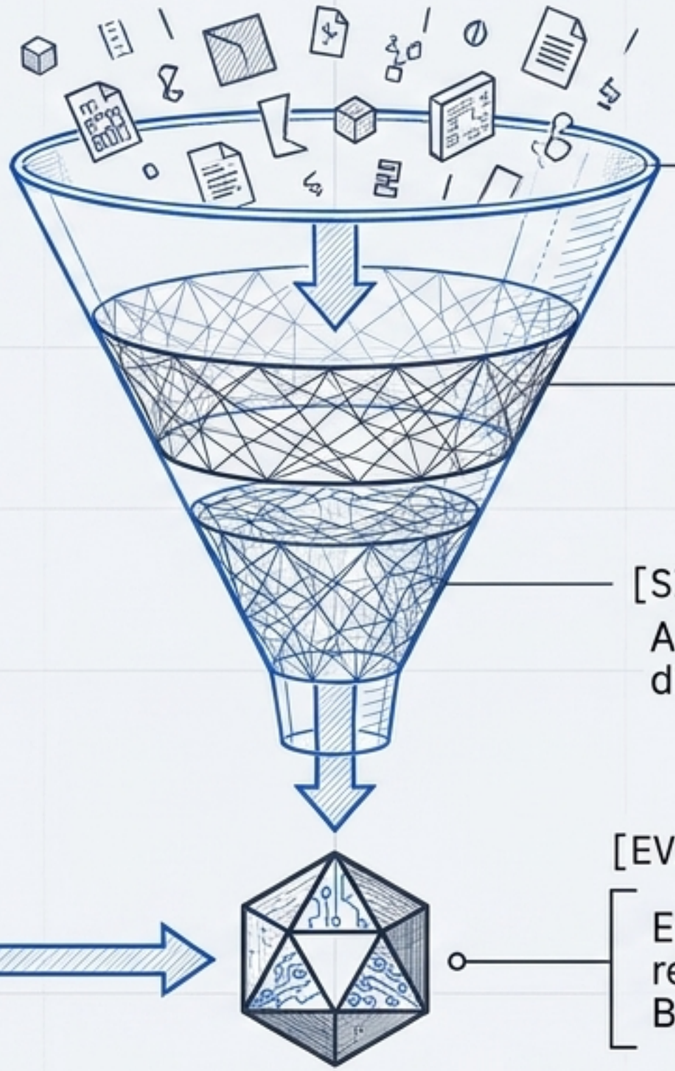
1. Agent actions pass into the sovereign Go Orchestrator.
2. Data is hashed and linked into a Merkle audit chain, enforcing partial IFC lattice rules.
3. Findings are permanently locked into a tamper-proof SQLite Write-Ahead Log (WAL) at the edge.

> If it isn't in the ledger, it didn't happen.

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Loop Stage 4: Scientific Synthesis



[RESULTS]

[AUTORESEARCH_ENGINE]

Organizes raw data into interactive experiment notebooks.

[REFUTED-GENOME_TRANSFER_PRIORS]

4-tier failure taxonomy with global intrinsic blacklisting.

[SINGLE-TAPE_AUTODIFF]

Analytical mixed-state parameter-shift differentiation via PennyLane.

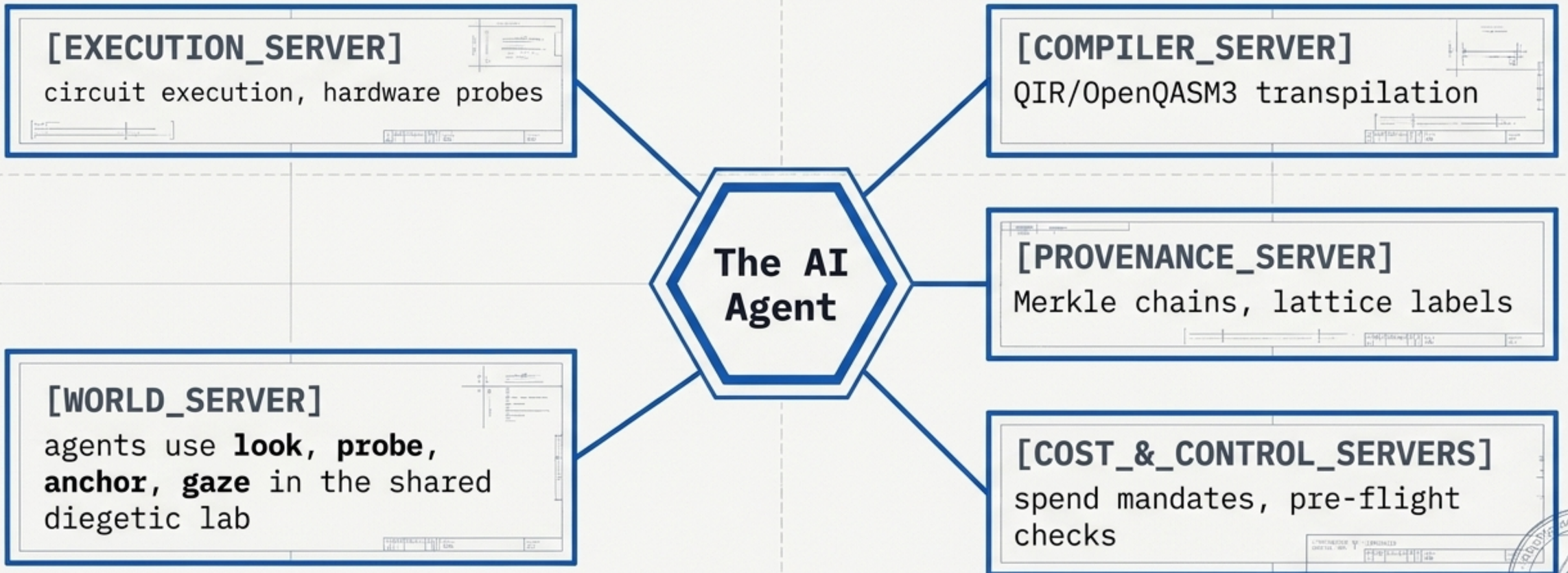
[EVALUATOR_GUIDED]

Evaluator-guided circuit co-design resulting in standardly-surprised, Brier-scored discoveries.

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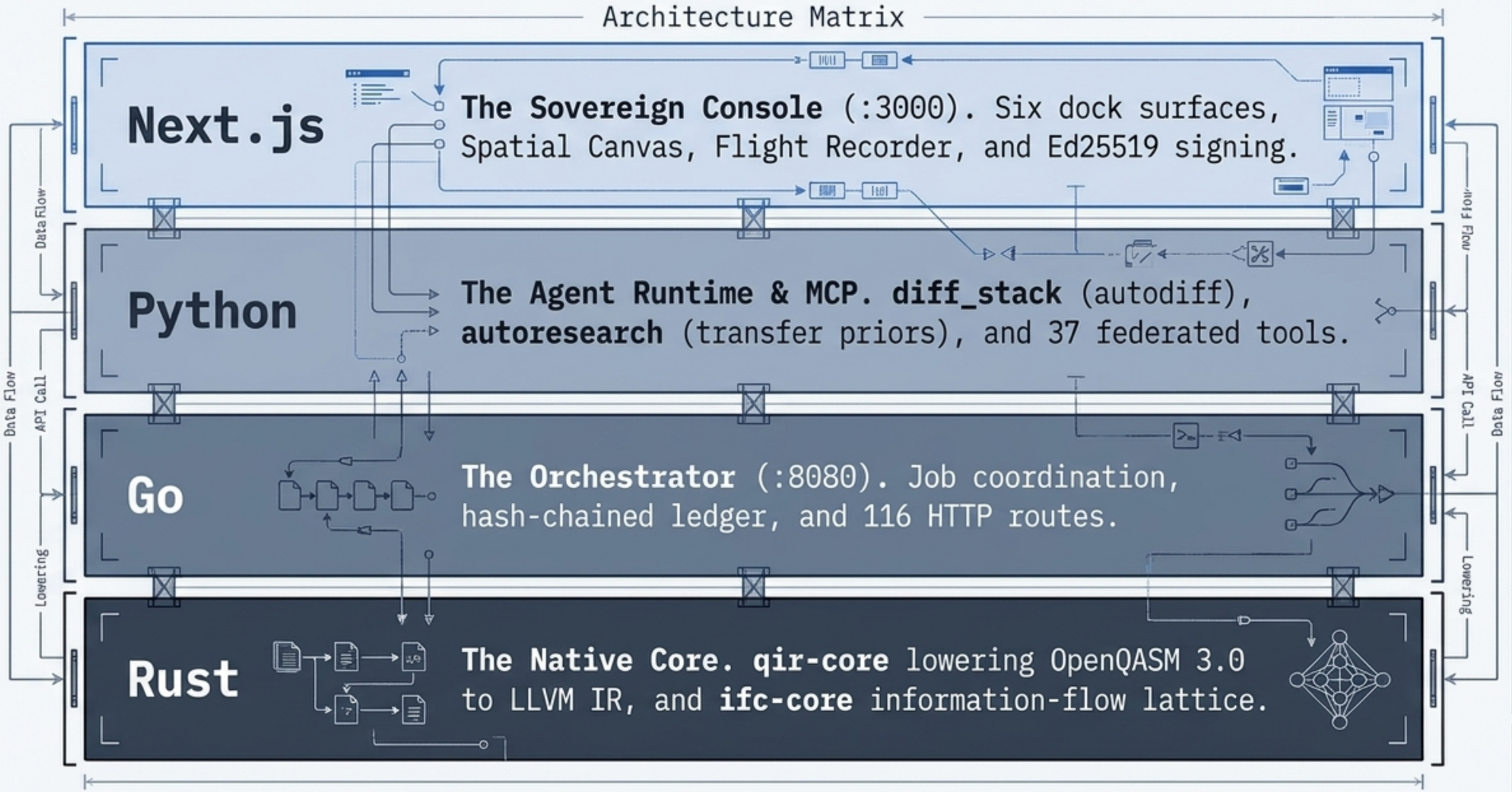
The Engine of Autonomy: Federated MCP Layer

Agents interface with the platform via 6 federated Python servers executing 37 deterministic tools over stdio.



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Polyglot System Topology



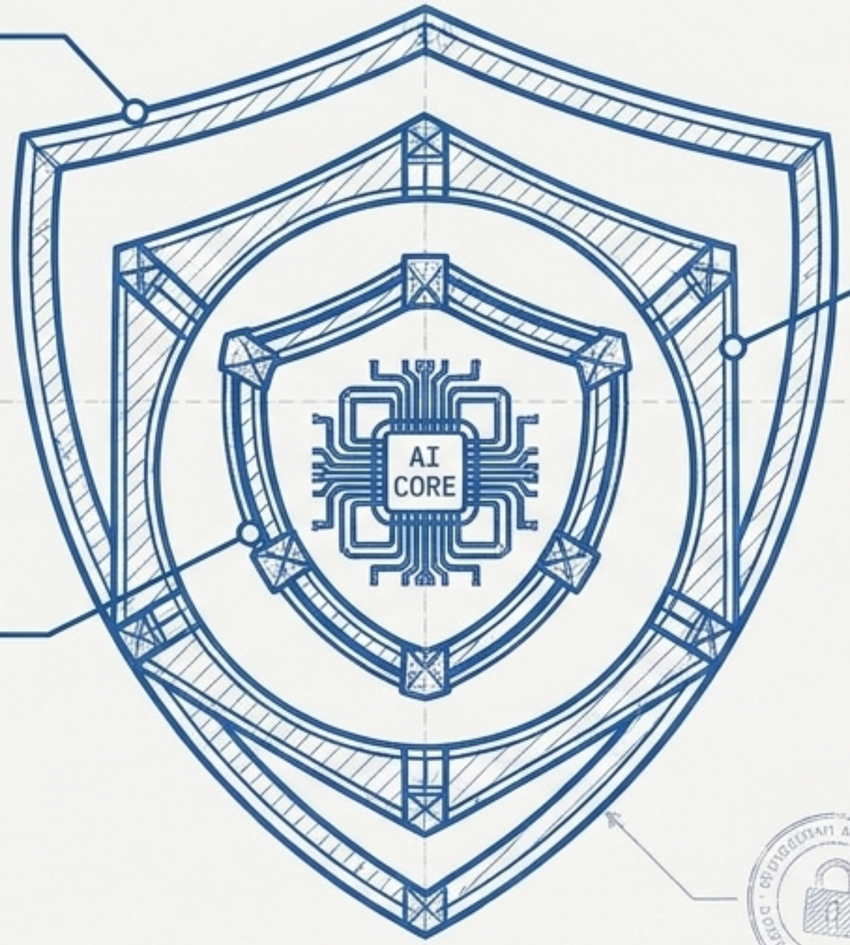
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Containment & Governance Lattice

Security Lattice

[Pre-Flight Control]

Strict spend mandates and budget checks. Operator approval cards required for execution.



[Information Flow]

3-Seam IFC (Information Flow Control) Lattice managing query fltering and cloud admission.

Fail-closed refusal of unmodelable usage.

[Process Containment]

Docker hardening with dropped capabilities and process-group **killpg**.

(Note: Python agent-code sandbox is actively escapable; **runsc** is opt-in).

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Settled Science & Technical Invariants



Single QIR Dialect

Standardized strictly on native Rust **qir-core** lowering directly to LLVM IR, parsing OpenQASM 3.0.



Single-Tape Autodiff

Joint analytical mixed-state parameter-shift differentiation of gate angles and channel noise on a single graph (via **PennyLane default.mixed**).



Dual-Model Packing

Employs **safe_288** (guaranteeing 0.000% overflow for production) paired with a **paper_256** tripwire accounting model for research replication.

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Local Execution & Zero-QPU Validation

Launch Go Orchestrator

```
cd platform/core
go build -o /tmp/kubonad ./cmd/kubonad
/tmp/kubonad -port 8080
```



Launch Next.js Console

```
cd platform/console
NEXT_PUBLIC_KUBONA_CONSOLE_TOKEN=dev-token
 pnpm dev
```

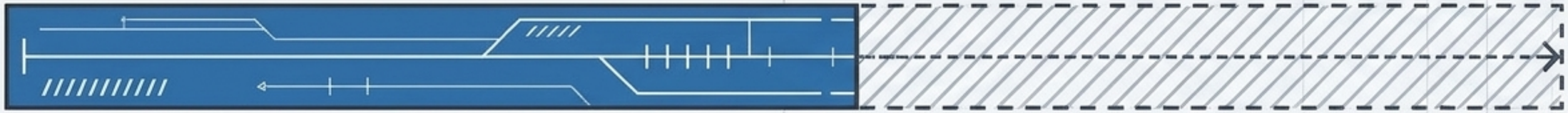
Run all 10 adopted local-simulation scenarios—including the Epistemic Tribunal, Flight Recorder, and Co-pilot Chat—without requiring physical quantum hardware.

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The Implementation Roadmap

Current State: Phases 0–5

Future Target: Phase 6



Quantum Core Foundations. Federated MCP servers, QIR native lowering, cost governance, immutable ledger, and autoresearch engine.

Science Discovery & Synthesis. Multi-user lab identities, workflow authoring studios, localized model training, and scholarly data integr.

WARNING: Active Research Platform. APIs, agent protocols, and the sovereign console evolve rapidly. Today’s green gate does not promise tomorrow’s.

Gemini Notebook

Open Source & Provenance

ATTRIBUTION DATA

AUTHORED BY: Mainza Kangombe / Milimo Quantum.

SOFTWARE LICENSE: Apache 2.0.

DOCUMENTATION LICENSE: CC-BY-4.0.

ACCESS PROTOCOLS

REPOSITORY LINK

github.com/mainza-ai/kubona

```
> git clone https://github.com/mainza-ai/kubona.git
```

Reference Note: Search 'Architecting the AI Quantum Bridge Inside the Kubona Stack' on YouTube for the technical deep dive.

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